Google Satellite Embedding V1 - small areas (2017-2024)

Abstract

This report describes the development, methodology, and validation of the Imago Embedding data product, a UK-wide dataset of satellite-derived embedding vectors aggregated to small-area statistical geographies across the United Kingdom (UK), including Lower Layer Super Output Areas (England and Wales), Data Zones (Scotland), and Super Output Areas (Northern Ireland), for the years 2017–2024. The data product is derived from the AlphaEarth Foundations Satellite Embedding Dataset V1. The workflow consists of two automated pipelines: a Google Earth Engine export pipeline that retrieves and tiles 64-dimensional Embeddings across British National Grid tiles, and aggregation pipeline that computes mean embedding vectors for each of the small statistical areas in the UK. The resulting GeoPackage output provides 64-dimensional embedding vectors (range [–1, 1]) for each small area statistical geography and year. Validation confirms 100% range compliance across ~24 million values and near-perfect correlation (mean r = 0.997) without systematic bias. The data product is published via the Imago Data Catalogue and is intended to support small-area analysis, spatial planning, urban research, and policy applications.

Keywords

Satellite embeddings; Earth observation; Small area statistical geography; Health and Wellbeing; Sustainability; Google embeddings; LSOA; UK

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1. Introduction

1.1 Background

Satellite-derived embeddings represent a new approach in geospatial science that summarises complex patterns contained within Earth observation imagery into compact numerical representations. These representations capture information about the physical and environmental characteristics of places while substantially reducing data complexity. The AlphaEarth Foundations Satellite Embedding Dataset V1, produced by Google **DeepMind** and available on Google Earth Engine, provides 64-dimensional, L2-normalised embedding vectors at 10-metre spatial resolution derived from annual satellite composites for 2017-2024. These embeddings capture rich environmental and structural information latent in satellite imagery, providing a foundation for a wide range of downstream analytical tasks, including classification, clustering, similarlity analysis, environmental characterisation, and predictive modelling.

Rather than representing specific land-cover classes, the embedding dimensions encode complex combinations of spectral, environmental, and built-environment characteristics learned from large volumes of satellite imagery. Areas with similar environmental and physical characteristics therefore tend to exhibit similar embedding representations, even when individual dimensions may not have direct physical interpretations. However, direct use of pixel-level embedding data at national scale is computationally demanding for most research and policy applications. This data product aims to aggregate these embeddings to small-area statistical geographies across the United Kingdom, including Lower Layer Super Output Areas (England and Wales), Data Zones (Scotland), and Super Output Areas (Northern Ireland), making them more usable for social science, urban analysis, and policy research.

1.2 Motivation

Small-area statistical geographies form the foundation of a large proportion of UK government statistics and policy analysis. They are widely used in the Census, the Index of Multiple Deprivation, public health monitoring, housing analysis, transport planning, and environmental assessment. Aggregating satellite-derived embeddings to these established geographies creates opportunities to integrate Earth observation-derived information with existing socioeconomic, demographic, and environmental datasets. Although there is a clear potential of satellite embeddings for characterising local environments, there is currently no publicly available data product which aggregates AlphaEarth satellite embeddings to UK small-area statistical geographies across the full 2017–2024 temporal range.

Because AlphaEarth embeddings are continuous latent representations normalised to a common feature space, arithmetic averaging provides a practical method for summarising neighbourhood-scale characteristics while preserving broad similarity relationships between areas. This enables embeddings to be analysed using conventional statistical and machine learning approaches while substantially reducing computational requirements compared with pixel-level datasets, thereby making the information more accessible for research, planning, and policy applications.

1.3 Objectives

This data product and the accompanying technical report have the following objectives:

- To develop a reproducible, automated pipeline for extracting and processing annual satellite embedding data from Google Earth Engine **(GEE)** at UK national scale.
- To aggregate pixel-level embedding vectors to UK small-area statistical geographies using a tile-aware weighted averaging procedure.
- To validate the resulting data product against direct Earth Engine extraction and confirm compliance with the theoretical embedding data value range.
- To publish the validated dataset via the Imago Data Catalogue in an accessible, well-documented format for use by researchers and policy professionals.

1.4 Contributions

The principal contributions of this work are:

- A UK-wide annual embedding dataset providing 64-dimensional satellite-derived embedding vectors for small-area statistical geographies across England, Wales, Scotland, and Northern Ireland between 2017 and 2024.
- A reproducible aggregation framework implementing pixel-weighted, multi-tile averaging for transforming pixel-level embedding data into neighbourhood-scale representations.
- A quantitative validation framework comparing pipeline outputs against direct Earth Engine extraction.
- A command-line tool for batch export of annual Google Earth Engine image collections to tiled GeoTIFF format.
- A scalable processing workflow that can be adapted to alternative administrative, statistical, or custom geographies.

For further details on the original Embedding dataset, see this blogpost.

2. Source Data and Study Area

2.1 Original Google Satellite Embedding Dataset

The source data for this product is the AlphaEarth Foundations Satellite Embedding Dataset V1, available on Google Earth Engine. This dataset provides annual 64-band raster dataset representing L2-normalised embedding vectors at 10-metre spatial resolution, with values bounded within [−1, 1]. The collection spans 2017 to 2024. An alternative distribution of the same underlying embeddings is available via Source Cooperative. In that product, AlphaEarth Foundation Embeddings are stored in signed 8-bit data type through nonlinear scaling, with −128 as the nodata value and the spatial index recorded within the filenames. This substantially reduces storage requirements but introduces a loss of numerical precision compared with the original Google Earth Engine distribution (see Section 3.4 and Appendix E).

All processing undertaken for this data product utilised the AlphaEarth Foundations dataset from Google Earth Engine, which represented the most recent version available at the time of processing.

2.2 Geographic Coverage and Aggregation Units

The target geographic unit for aggregation is the set of official small-area statistical geographies used across the UK, comprising Lower Layer Super Output Area (LSOA), the standard small-area statistical geography for England and Wales, Data Zones in Scotland, and Super Output Areas in Northern Ireland. The resulting dataset covers 46,844 small-area statistical geographies annually between 2017 and 2024 and is from the Imago Data Catalogue.

2.3 Input Data Specifications

The input data as received from Google Earth Engine has the following characteristics:

- **Format:** GeoTIFF, optionally with COG (Cloud-Optimised GeoTIFF) layout.
- **Bands:** 64 bands, labelled A00 to A63.
- **Data type:** Int16, LZW compression.
- **CRS:** UTM zone projection
- **Spatial resolution:** 10 m
- **NoData:** The exported embedding tiles contain valid embedding values throughout the image extent and do not define a separate NoData value.
- **Tile edge artefact:** Exported tiles may contain one additional pixel along the northern and eastern edges due to pixel alignment and boundary handling during export (e.g., expected 2000 × 2000 pixels yields 2001 × 2001 pixels in output).

3. Methodology

3.1 Workflow Overview

The IMAGO processing pipeline consists of three sequential processing stages (*Figure 1*):

1. **Data Extraction** – annual AlphaEarth embedding data are retrieved from Google Earth Engine and exported as GeoTIFF imagery covering the area of interest.
2. **Data Preparation** – exported imagery is reprojected to the British National Grid, organised into a standardised tiling structure, and transformed from floating-point values to a scaled Int16 representation to improve storage efficiency.

3. **Small-Area Aggregation** – embedding values are aggregated from pixel level to small-area statistical geographies using zonal statistics and a tile-aware weighted averaging procedure.

The final output is a GeoPackage containing annual embedding vectors for all small-area statistical geographies across the United Kingdom. These stages are described in Sections 3.2–3.4.

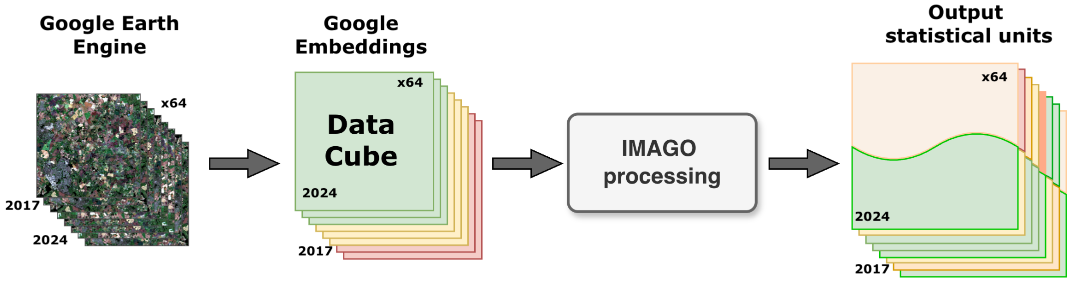

Figure 1: Conceptual flowchart illustrating the main processing steps

3.2 Data Extraction

A command-line tool has been developed to access annual satellite collections on Google Earth Engine, filter by the area and timeframe of interest, and export in GeoTIFF format to Google Cloud Storage. This tool can also be used to export tiles to Google Drive.

Main requirements are:

- Polygonal area of interest, which can be tiled (for example, the UK gridded by tiles of 20 × 20 km)
- Year of interest

The pipeline accesses the AlphaEarth embedding collection on Google Earth Engine, filters the data according to the specified area and year of interest, and exports the resulting imagery in batches through the Google Earth Engine API (*Figure 2*).

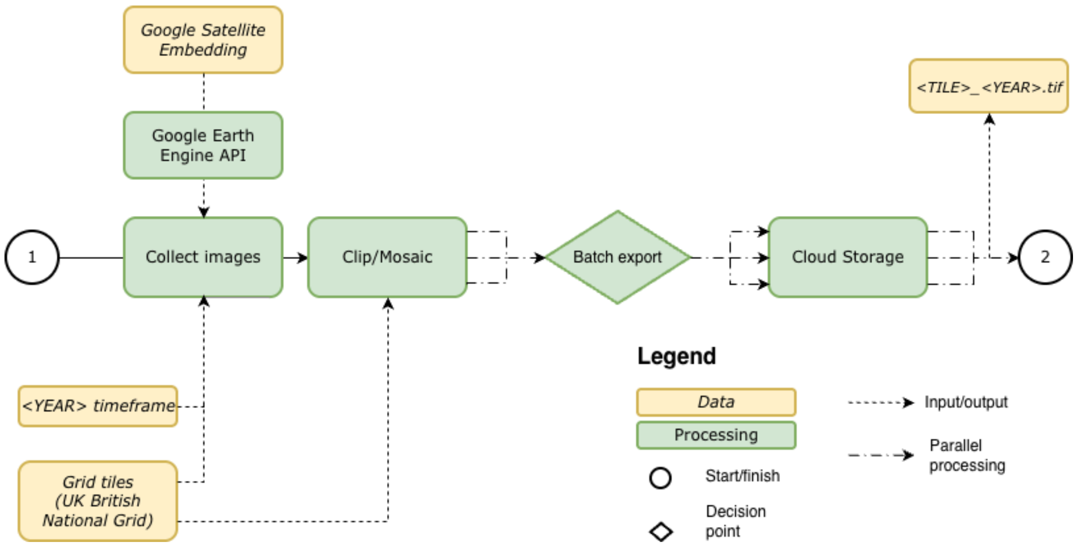

Figure 2: Conceptual flowchart illustrating the input data, input/output operations, processing steps and output data in the Google Earth Engine download pipeline

In this pipeline, batch export tasks are used — `Export.image.toDrive` or `Export.image.toCloudStorage` — which produce identical outputs.

Full CLI usage is documented in Appendix A.

3.3 Data Preparation

3.3.1 Tiling Strategy

The Imago intermediate dataset in GeoTIFF format is gridded using 20 × 20 km British National Grid (BNG) tiles, resulting in 858 tiles for the full UK coverage. This tiling structure aligns with the OSGB36/British National Grid (EPSG:27700) coordinate reference system used throughout the intermediate processing stage. In contrast, the original AlphaEarth embedding data covering the UK are distributed across 39 larger tiles within GEE. During the data preparation stage, these tiles are reprojected and regridded into the standaised BNG tile structure used throughout the remainder of the workflow.

3.3.2 Embedding Transformation and Scaling

To reduce storage requirements while preserving precision, the pipeline optionally scales the native `float64` embedding values (range [−1, +1]) to `Int16` representation (range [−32767, +32767]), reducing output size by approximately one third.

To recover original decimal values (to "descale"), apply:

$$x_{\text{original}} = \frac{x_{\text{int16}}}{32767}$$

The scaled Int16 representation preserves the original embedding values with very high precision (Figure 3). Validation using tile SJ26 (2024) indicates that scaling errors are centred around zero and exhibit no evidence of systematic bias. The mean error and RMSE occur at approximately the sixth decimal place, while the maximum absolute error is approximately 1.6×10^{-5}, corresponding to the fifth decimal place. These results demonstrate that the Int16 representation provides substantial storage savings while maintaining effectively lossless precision for downstream analysis. Full scaling precision statistics are provided in Appendix E.

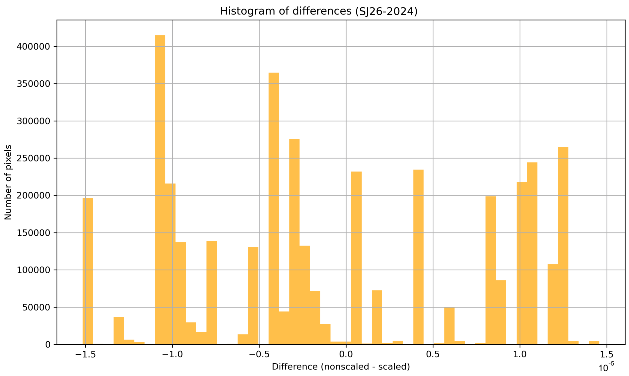

Figure 3: Distribution of scaling errors between the original floating-point embedding values and the corresponding Int16-scaled representation for embedding dimension A00, Tile SJ26 (2024).

3.3.3 Intermediate Data Specifications

After reprojection and scaling, the intermediate data has the following characteristics:

- **Format:** GeoTIFF (without COG layout).
- **Bands:** 64 bands, labelled A00 to A63.
- **Data type:** Int16, LZW compression.
- **CRS:** EPSG:27700, OSGB36/British National Grid; spatial resolution 10 m.
- **NoData:** Undefined, with one extra pixel along the northern and eastern tile edges. Pixels equal to 0 are filtered out during further processing to avoid calculation distortions.
- **Tiling:** Gridded by 20 × 20 km National Grid tiles.
- **Value range:** –32767 to +32767 (scaled); equivalent to [–1, +1] in original float64 space.

3.4 Small Area Aggregation

3.4.1 LSOA Aggregation Pipeline

The aggregation pipeline converts tile-based embedding datasets into small-area statistical geography summaries. For each year, the workflow identifies all tiles intersecting a geographic unit, extracts the corresponding embedding values, and computes mean values for each of the 64 embedding dimensions using zonal statistics. The output is a GeoPackage containing one record per geographic unit and 64 corresponding embedding attributes. To facilitate processing of multiple years, the workflow uses a set of configuration templates and automation scripts (Figure 4). A master script, `run_all_years.sh`, generates a year-specific configuration file (`config_$YEAR.yaml`) and execution script (`run_$YEAR.sh`) for each year of interest (2017–2024). The configuration file stores all processing parameters, including input and output locations, memory allocation, and parallel processing settings, while the execution script launches the aggregation workflow using these settings.

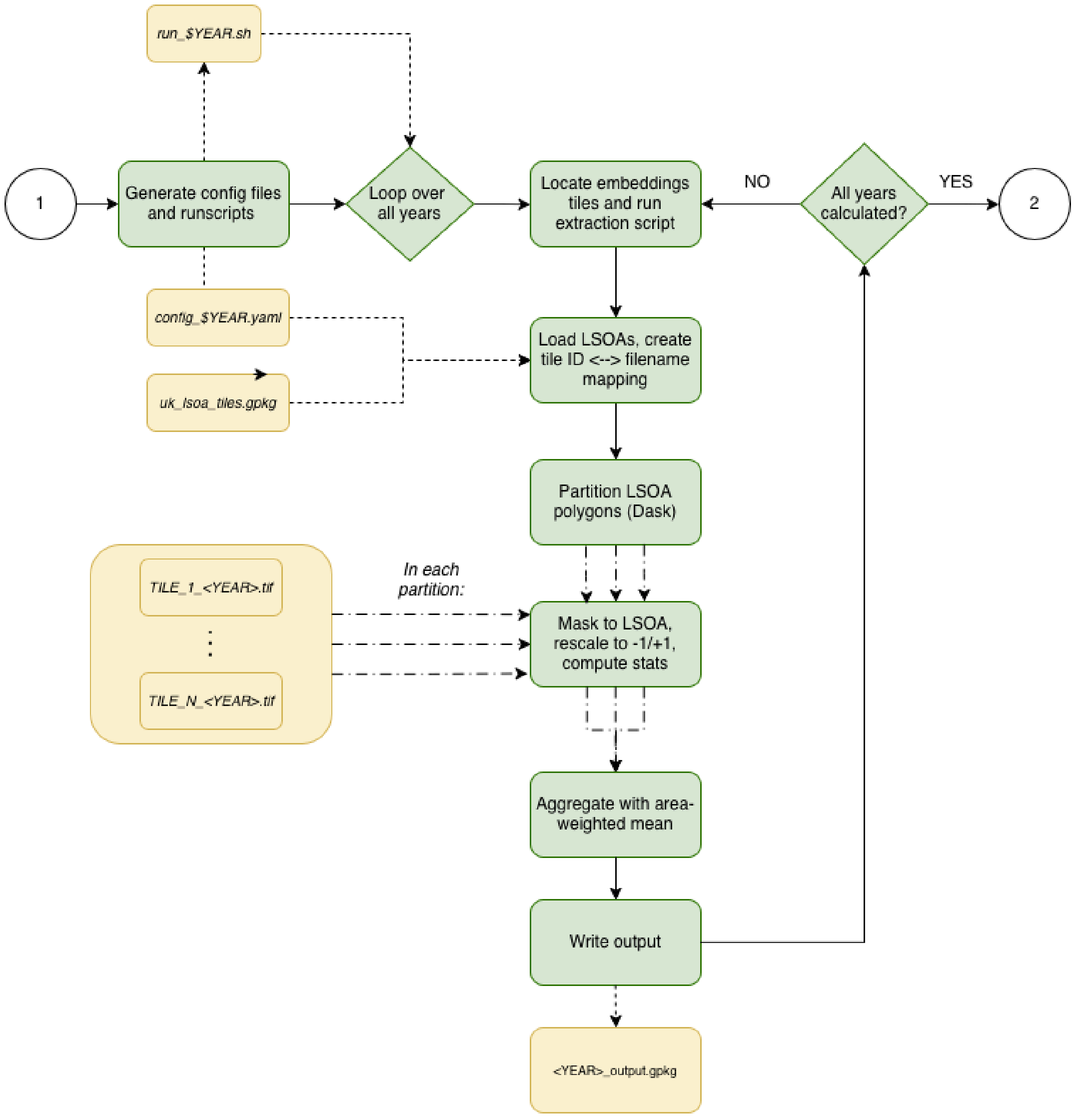

Figure 4: Conceptual flowchart illustrating the input data, input/output operations, processing steps and output data in the LSOA aggregation pipeline

The aggregation itself is performed by `embed_to_lsoa.py`, which accesses routines from the IMAGO toolkit repository. The workflow first establishes the relationship between geographic units and the tiles that intersect them, creating a tile-to-polygon mapping that identifies the relevant embedding tiles for each LSOA. The required tiles are then loaded and processed using zonal statistics to calculate mean embedding values for each of the 64 embedding dimensions across all pixels falling within the geographic unit boundary. To support national-scale processing, calculations are parallelised using Dask, with geographic units processed in chunks across multiple workers, while `rasterio` performs the underlying raster operations and zonal statistics calculations. This enables efficient processing of large embedding datasets while maintaining a relatively modest computational footprint.

Because the workflow relies solely on a geographic boundary dataset as input, the same aggregation procedure can readily be applied to alternative geographies, including Middle Layer Super Output Areas (MSOAs), local authority districts, Data Zones, Super Output Areas, or custom administrative boundaries, provided that an appropriate tile-to-polygon mapping can be established.

3.4.2 Weighted Averaging Procedure for Multi-Tile Geographies

For LSOAs that span multiple tiles, a tile-aware weighted averaging procedure is applied:

1. The mean embedding value is calculated per tile for each LSOA-tile intersection.
2. The number of overlapping pixels within each tile is recorded alongside the computed their computed mean.
3. The final small area geography embedding vector is computed by averaging the per-tile mean embedding values using the number of overlapping pixels in each tile as weights.

For embedding dimension (j), the weighted average is calculated as:

$$E_j = \frac{\sum_{i=1}^{n} p_i E_{ij}}{\sum_{i=1}^{n} p_i}$$

where E_j denotes the final LSOA-level embedding value for dimension j, E_{ij} denotes the mean embedding value for dimension j calculated within tile i, p_i denotes the number of overlapping pixels within tile i, and n denotes the number of intersecting tiles.

This approach ensures that each tile contributes to the final embedding vector in proportion to its spatial contribution to the geographic unit being observed. This approach is necessary because some LSOAs, particularly in areas near tile boundaries, may cover two or more 20 × 20 km tiles.

The current implementation uses `rasterio` for zonal statistics, which is faster than libraries such as `exactextract` but treats all overlapping pixels as contributing equally and does not apply fractional weighting to partially intersected boundary pixels. While this may introduce minor uncertainty along geographic boundaries, the effect is expected to be limited given the 10-metre spatial resolution of the source data. The implications of this assumption are examined in the validation analysis presented in Section 5.

3.5 Computational Infrastructure and Parallel Processing

3.5.1 GEE Export Pipeline

Processing is performed on Google Earth Engine infrastructure. Two distinct time concepts are relevant:

1. **Python pipeline runtime** (client wall-clock time) — time on the client machine between two Python timestamps. Does not reflect billing or quota consumption.
2. **EECU (Earth Engine Compute Units)** — compute consumption across all parallel workers; reflects billing and quotas. Includes I/O reads and array manipulations, but excludes latency, queue scheduling, client waiting, and Google Drive/Cloud upload time.

Python pipeline runtime is the sum of individual task runtimes plus client-side preparation, initialisation, and scheduling time. Even identical requests can be processed in very different times (see GEE Computation Overview). EECU time for the same area of interest has been observed to vary by a factor of 2.8, while total runtime can vary by up to a factor of 8.8. For detailed benchmarking tables, please see the Appendix B.

3.5.2 LSOA Aggregation Pipeline

The LSOA aggregation step is run on a virtual machine (VM) with 8 CPUs and 32 GB of memory, chosen as a balance between performance and cost. Data is accessed from Google Cloud Storage (GCS) via bucket mounting to the VM, avoiding the need to download the complete dataset locally. This introduces I/O overhead compared to local disk access, but avoids the need to download the full dataset locally.

Key implementation choices for performance:

- **Chunked band processing:** Calculation is split into groups of bands (e.g., 8 bands per chunk, 8 chunks for 64 bands total), enabling processing without spilling to disk.
- **LRU caching:** `functools.lru_cache` is used to cache tile metadata from repeated `rasterio.open` calls, improving overall performance by approximately 25%.
- **Dask parallelisation:** `dask` is used to parallelise over LSOAs in chunks, with 8 concurrent workers.

Using this configuration, the aggregation of a single annual dataset requires approximately 1.5 hours on average. Consequently, the complete 2017–2024 time series can typically be processed within approximately one day. Detailed benchmarking results and performance statistics are provided in Appendix B.

4. Data Product Description

4.1 Output Structure

The output data product is distributed as eight annual GeoPackage (.gpkg) files, one for each year between 2017 and 2024. Each file contains one row per small-area statistical geography and the corresponding embedding attributes. Each row contains the geometry of the statistical geography (from 2021 LSOA boundaries) and 64 embedding attributes. The product covers 46,844 small-area statistical geographies annually across England, Wales, Scotland, and Northern Ireland.

4.2 Embedding Dimensions

Each small-area statistical geography includes 64 embedding dimensions (band1_mean to band64_mean), each representing the mean embedding value across all pixels within the geographic unit for that dimension and year. These are L2-normalised vectors derived from the AlphaEarth Foundation model, encoding spectral and structural information from annual satellite composites.

4.3 File Formats and Storage

The main output format is GeoPackage (`.gpkg`). During processing, tiling, compression, and Int16 scaling were used to reduce storage requirements for the intermediate embedding datasets. These procedures reduced the total intermediate data volume from an estimated ~4 TiB (unscaled floating-point values) to approximately 2.02 TiB (Int16, LZW-compressed) across all eight years.

4.4 Metadata and Attribute Schema

The output GeoPackage includes:

- A unique statistical geography identifier.
- Polygon geometry representing the statistical geography boundary.
- 64 embedding attributes, each storing the mean embedding value for a single embedding dimension.

All embedding attributes are stored using double-precision floating-point data types.

Table 1 summarises the principal fields included in the GeoPackage.

Field Type	Description
Geography ID	Unique identifier for the statistical geography
Geometry	Polygon boundary geometry
band1_mean – band64_mean	Mean embedding values for the 64 embedding dimensions

Additional summary statistics, including median, minimum, maximum, standard deviation, and sum, can be generated through the aggregation pipeline if required. Full field descriptions are provided in Appendix D.

5. Technical Validation

5.1 Validation Framework

Validation of the small-area statistical geography embedding datasets was conducted through two primary checks:

1. **Range validation** — verifying that all embedding values fall within the theoretical [–1, 1] bounds.
2. **Random sampling validation** — comparing pipeline-generated mean embedding values against values extracted directly from Google Earth Engine for a random sample of geographic unit-year pairs.

Together, these validation procedures assess both compliance with the theoretical properties of the AlphaEarth embeddings and the ability of the aggregation workflow to reproduce values obtained directly from Google Earth Engine (GEE).

Validation Task	Description	Status
Range Check	Verify all embedding values are within [–1, 1]	✅ PASSED
Random Sampling Validation	Compare pipeline with EE extraction (100 samples)	✅ PASSED

5.2 Range Validation

The AlphaEarth embedding model produces values theoretically bounded within [–1, 1]. This check systematically inspected every numeric value across all 64 embedding dimensions for each year from 2017 to 2024.

- Total values inspected per year: 46,844 geographic units × 64 dimensions = 2,998,016 values
- Total values inspected across 8 years: ~24 million values

Year	LSOAs	Dimensions	Total Values	Out-of-Range	Status
2017	46,844	64	2,998,016	0	✅
2018	46,844	64	2,998,016	0	✅
2019	46,844	64	2,998,016	0	✅
2020	46,844	64	2,998,016	0	✅
2021	46,844	64	2,998,016	0	✅
2022	46,844	64	2,998,016	0	✅
2023	46,844	64	2,998,016	0	✅
2024	46,844	64	2,998,016	0	✅

No out-of-range values were detected in any year or embedding dimension. Consequently, 100% compliance with the theoretical [−1, 1] constraint was achieved across all 23,984,128 inspected values. These results confirm that the aggregation, scaling, and export procedures preserve the theoretical properties of the original AlphaEarth embeddings and do not introduce numerical artefacts that violate the expected value range.

5.3 Random Sampling

A random sample of 100 (LSOA, year) pairs was drawn from 400 available pairs (50 LSOAs across eight years). For each sampled pair, the pipeline-generated mean embedding vector was compared against the mean embedding vector computed by direct extraction from Google Earth Engine (GEE). The validation sample was designed to provide coverage across the full 2017–2024 period while maintaining computational feasibility for direct GEE extraction. Validation metrics included mean difference, Pearson correlation, and the standard deviation of differences. Conservative quality-assurance thresholds of ±0.1 for mean differences and ($r > 0.5$) for correlations were adopted. As demonstrated in the following subsections, the observed results substantially exceed these minimum requirements.

5.3.1 Statistical Consistency and Correlation Assessment

Metric	Value
Total comparisons completed	100
Unique LSOAs sampled	42
Years covered	2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024

Mean difference (Pipeline – EE)	−0.000049
Std deviation of differences	0.000698
Mean correlation	0.9973
Min correlation	0.9859
Max correlation	1.0000

Criterion	Threshold	Pass Rate
Mean values within tolerance	±0.1	100.0%
Correlation above threshold	r > 0.5	100.0%

Agreement between the pipeline-generated values and direct Google Earth Engine extraction was extremely high. Mean differences were negligible, correlations consistently approached unity, and all sampled observations exceeded the predefined validation criteria.

5.3.2 Systematic Bias Testing

Test	Value	Result
t-statistic	−0.7085	
p-value	0.4803	✅ PASS
Conclusion		No systematic bias detected

The mean difference between pipeline-derived values and direct Google Earth Engine extraction was not significantly different from zero ((t = -0.7085), (p = 0.4803)), providing no evidence of systematic bias within the aggregation workflow.

5.3.3 Temporal Consistency

The yearly breakdown of mean differences and standard deviations confirms stability across the full 2017–2024 period, with no evidence of temporal drift:

Year	Count	Mean Difference	Std Deviation
2017	11	−0.000027	0.000713
2018	13	−0.000176	0.000716
2019	13	0.000040	0.000887
2020	15	0.000082	0.000602

2021	10	0.000120	0.000648
2022	10	−0.000426	0.000538
2023	15	−0.000018	0.000821
2024	13	−0.000062	0.000605

Mean annual differences ranged from −0.000426 to 0.000120, while annual standard deviations ranged from 0.000538 to 0.000887. Mean differences remain tightly centred around zero across all years, with consistently negligible variation and no evidence of temporal drift.

5.4 Visual Validation

Figure 5 provides visual confirmation of the statistical validation results presented in Sections 5.2 and 5.3. Together, the visualisations demonstrate strong agreement between pipeline-generated values and direct Google Earth Engine extraction, with no evidence of systematic bias or temporal degradation.

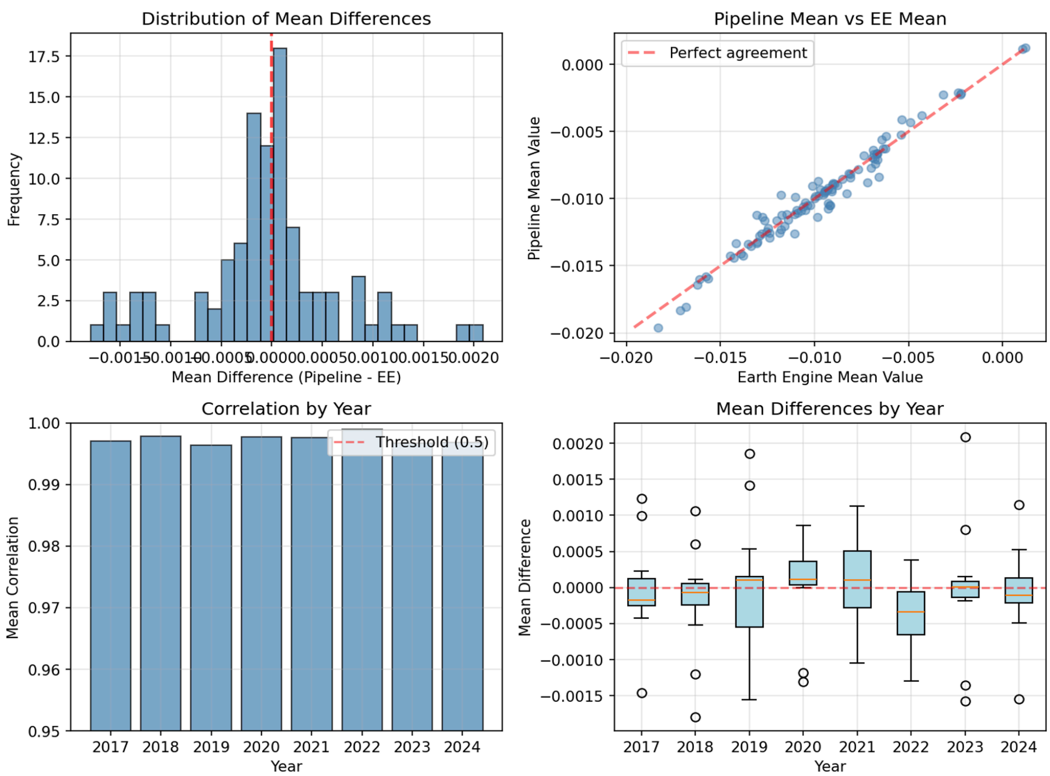

Figure 5: Validation visualisations showing distribution of mean differences, correlation between pipeline and Earth Engine data, and yearly performance.

Top-Left Panel — Distribution of Mean Differences: The distribution is tightly centred around zero with no visible skew, indicating neither dataset systematically overestimates or underestimates the other. All values cluster within ±0.002 — well inside the ±0.1 tolerance threshold.

Top-Right Panel — Pipeline Mean vs. EE Mean: Near-perfect alignment of points along the identity line demonstrates that pipeline values are almost identical to direct Earth Engine extraction across the full value range.

Bottom-Left Panel — Correlation by Year: Correlations consistently exceed 0.99 across all eight years (well above the 0.5 threshold), demonstrating stable data quality and extraction fidelity with no temporal degradation.

Bottom-Right Panel — Mean Differences by Year: Medians cluster tightly around zero with extremely narrow interquartile ranges, confirming no single year exhibits systematic bias.

Combined Validation Summary:

Validation Criterion	Result
Range check [−1, 1]	100% compliant
Statistical consistency with EE	100% pass rate
Systematic bias	None detected
Year-to-year consistency	Stable across all years

Collectively, the validation results demonstrate that the aggregation workflow reproduces Google Earth Engine outputs with high fidelity, preserves the theoretical properties of the AlphaEarth embeddings, and maintains stable performance across the full 2017–2024 period.

6. Performance Evaluation

6.1 Runtime and Compute Performance

Processing was undertaken using a combination of Google Earth Engine (GEE) and cloud-hosted virtual machine infrastructure. Detailed runtime and Earth Engine Compute Unit (EECU) statistics are provided in Appendix B.

The small-area aggregation step runs on a virtual machine with 8 CPUs and 32 GB of memory, accessing data directly from Google Cloud Storage. On average, processing a single year requires approximately 1.5 hours, enabling the complete 2017–2024 dataset to be processed within approximately one day. This demonstrates that national-scale aggregation of the AlphaEarth embeddings can be achieved using relatively modest cloud computing resources.

6.2 Storage Optimisation

The pipeline achieves significant storage reduction through three complementary strategies:

- **Tiling** (20 × 20 km BNG tiles), which enables parallel processing and avoids loading the full national dataset into memory.
- **Int16 scaling** (multiplying float64 values by 32,767), which reduces per-file size by approximately one third compared to float32 or float64 representation.
- **LZW compression**, applied automatically to all GeoTIFF outputs.

Together these reduce the total UK dataset from an estimated ~4 TiB (unscaled) to ~2.02 TiB across the full 2017–2024 period while preserving the information content of the original embeddings (see Appendix E).

6.3 Scalability and Reproducibility

The pipeline is designed to be fully reproducible and extensible:

- The aggregation pipeline accepts any input GeoPackage with an appropriate tile ↔ polygon mapping, enabling aggregation to other geographies. The same methodology can therefore be applied to small-area statistical geographies, Middle Layer Super Output Areas (MSOAs), local authority districts (LADs), wards, output areas, bespoke study areas, and other administrative or analytical geographies.
- All processing parameters are stored within configuration files, enabling automated year-by-year execution with minimal manual intervention and ensuring that identical inputs produce identical outputs. This supports reproducibility and facilitates the reprocessing of future AlphaEarth releases or alternative geographic boundary datasets.
- The workflow is inherently parallelisable because geographic units are processed independently. Consequently, runtime can be reduced through the allocation of additional computational resources, subject to available memory and storage constraints. Performance can be improved by increasing the number of Dask workers, provided sufficient virtual machine memory is available.

7. Usage Notes and Limitations

7.1 Intended Applications

The Imago Embedding data product is intended for use in:

- Small-area socioeconomic and environmental analysis using satellite-derived features.

- Machine learning and statistical modelling tasks requiring area-level environmental representations.
- Similarity analysis, clustering, and dimensionality reduction of neighbourhood-scale environments.
- Environmental and built-environment characterisation using latent representations derived from Earth observation imagery.
- Change detection and temporal trend analysis at small-area statistical geography level (2017–2024).
- Integration with administrative data (e.g., deprivation indices, Census data, health statistics, and environmental indicators)
- Research and policy applications requiring a spatially comprehensive, temporally consistent feature set for UK small-area statistical geographies.

7.2 Recommended Analytical Practices

- When using the Int16-scaled intermediate data, apply the descaling formula

$$x_{\text{original}} = \frac{x_{\text{int16}}}{32767}$$

  before analysis to recover the original [−1, 1] range.
- Embedding dimensions should generally be interpreted collectively rather than individually. Individual dimensions do not correspond to specific physical, environmental, or socioeconomic variables, and their analytical value arises primarily from the relationships among dimensions within the full embedding vector.
- For aggregation to other geographies (e.g., MSOA), ensure the input GeoPackage contains an appropriate tile ↔ polygon mapping consistent with the 20 × 20 km BNG tile grid.
- Temporal changes in embedding values should be interpreted as changes in the latent representation of the observed environment rather than direct measurements of specific environmental, land-cover, or socioeconomic changes.

7.3 Known Limitations

Several limitations should be considered when using the data product.

- **Partial-pixel boundary effects:** The current pipeline uses `rasterio` for zonal statistics, which treats all overlapping pixels as contributing equally (no partial-pixel weighting). This introduces a small positive bias in mean values for small, high-density geographies where the proportion of

boundary-intersecting pixels is higher. The 10-metre resolution substantially mitigates this effect, but it remains a known source of minor imprecision.
- **Random sampling validation scope:** The sampling validation was limited to 100 (geographic unit, year) pairs from 400 available, and focuses on statistical consistency rather than exact pixel-level matching. Different aggregation methods may produce slightly different values.
- **Data quality:** Although data quality issues affecting 2017 (related to Sentinel-1 image dropout) have been addressed in the current GEE collection, users should be aware of this history when using 2017 data.

8. Discussion

8.1 Scientific and Policy Implications

The Imago Embedding data product makes satellite-derived environmental representations accessible at the standard UK small-area geography for the first time at national scale and annual temporal resolution. This product helps bridge the gap between raw Earth observation data and administrative statistical geographies, enabling a new class of analyses that combine satellite-derived features with socioeconomic, health, housing, environmental, and demographic data available at the scale of small statistical areas. By providing annual, nationally consistent environmental representations, the data product creates opportunities for longitudinal analyses of neighbourhood change, environmental inequalities, urban development, and place-based policy interventions that would otherwise require substantial Earth observation expertise and computational resources.

The strong agreement between pipeline-derived values and direct Google Earth Engine extraction provides confidence that the product can be used in downstream statistical, machine-learning, and policy applications without introducing substantial aggregation artefacts.

8.2 Methodological Contributions

The workflow demonstrates a scalable approach for transforming national-scale Earth observation foundation model outputs into small-area statistical geography datasets suitable for research and policy applications. The tile-aware weighted averaging procedure for multi-tile geographic units ensures that areas intersecting tile boundaries receive spatially representative embedding values while maintaining computational efficiency at national scale. Similarly, the Int16 scaling strategy preserves embedding values with negligible numerical error while substantially reducing storage requirements, demonstrating a practical approach for managing large Earth observation datasets. Together with automated Google Earth Engine extraction and comprehensive validation, these components provide a reproducible framework that can be adapted to future embedding products and alternative geographic boundaries.

8.3 Future Development Opportunities

- **Expanded temporal coverage:** Extension of the pipeline to accommodate new years of the Google Satellite Embedding dataset as they become available.
- **Additional geographies:** Production of equivalent datasets for MSOAs, wards, local authority districts, and other administrative units.
- **Additional foundation models:** Extension of the workflow to alternative Earth observation embedding products and foundation models.
- **Temporal consistency checks:** Year-over-year plausibility checks and outlier detection for individual LSOA time series.
- **Derived products and analytical tools**: Development of dimensionality-reduced representations, similarity-search tools, and derived indicators to support wider adoption by research and policy communities.
- **Windowed reads:** Implementation of true windowed rasterio reads to reduce memory pressure and enable higher parallelism in the LSOA aggregation step.

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9. Data and Code Availability

9.1 Data Access

The validated small area statistical geography satellite embedding dataset (2017–2024) is published via the Imago Data Catalogue: https://data.imago.ac.uk/datasets/google-satellite-embedding-v1-small-areas-20

The catalogue entry includes the annual GeoPackage datasets, metadata, documentation, and supporting resources required for data use and interpretation.

9.2 Code and Workflow Availability

The source code is available on Imago Github repository.

The pipeline comprises two main scripts:

- `src/download_ee.py` — GEE export CLI tool (see Appendix A for full usage)
- `src/embed_to_lsoa.py` — small-area aggregation workflow aggregation pipeline

Supporting configuration files, templates, and workflow documentation are provided to facilitate reproducibility and adaptation to alternative geographies.

9.3 Licensing

10. Conclusion

This report has described the development, methodology, and validation of the Imago small-area statistical geography satellite embedding data product for the United Kingdom, covering 2017 to 2024. The product provides 64-dimensional, L2-normalised embedding vectors for 46,844 small-area statistical geographies annually, derived from the Google Satellite Embedding Dataset V1 through an automated workflow comprising Google Earth Engine data extraction, intermediate data preparation, and small-area aggregation. Validation confirmed 100% compliance with the theoretical [–1, 1] embedding value range across approximately 24 million values and demonstrated near-perfect agreement (mean (r = 0.997); mean difference = -4.95×10^{-5} with direct Google Earth Engine extraction. No evidence of systematic bias was detected, and workflow performance remained stable across the full 2017–2024 period.

The data product is published via the Imago Data Catalogue and is intended to support a wide range of small-area analytical, research, and policy applications. By transforming national-scale Earth observation foundation model outputs into accessible geographic-unit representations, the product helps bridge the gap between Earth observation data and administrative statistical geographies, creating new opportunities for environmental, socioeconomic, health, housing, and policy research. Future development may focus on enhanced boundary handling, additional embedding products and foundation models, derived analytical products, and automated validation workflows for future annual releases.

Acknowledgements

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References

Key sources:

- Google Satellite Embedding Dataset V1: https://developers.google.com/earth-engine/datasets/catalog/GOOGLE_SATELLITE_EMBEDDING_V1_ANNUAL
- Google Earth Engine blog post on the Satellite Embedding Dataset: https://medium.com/google-earth/ai-powered-pixels-introducing-googles-satellite-embedding-dataset-31744c1f4650
- Source Cooperative AlphaEarth Foundation Embeddings: https://source.coop/tge-labs/aef
- GEE Computation Overview (stability and predictability): https://developers.google.com/earth-engine/guides/computation_overview#stability_and_predictability
- London Datastore statistical GIS boundary files: https://data.london.gov.uk/dataset/statistical-gis-boundary-files-for-london-20od9/

Appendices

Appendix A: Command-Line Tool Configuration and Usage Examples

The GEE export pipeline is invoked via:

```
python src/download_ee.py
```

Additional help and usage examples:

```
python src/download_ee.py --help
```

Positional arguments are not used; the CLI uses named options only. The following options are available:

Option	Description	Default
`--collection`	Earth Engine image collection to extract	`GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL`
`--project`, `-p`	Google Earth Engine project ID	`imago`

--auth-mode	Earth Engine authentication mode	gcloud
--tiles	Path to the gridded area of interest (tiles)	data/uk_1km_grid_sample1.gpkg
--storage	Export destination – Google Cloud Storage or Google Drive	cloud
--folder	Cloud Storage bucket or Google Drive folder	embed2social-storage
--year, -y	Year to extract dataset for	2024
--verbose, -v	Enable verbose logging (WARNING: can produce very large logs)	False
--cog	Export as Cloud Optimised GeoTIFF (COG)	False
--crs	Coordinate Reference System (CRS) of the output	EPSG:27700
--res	Spatial resolution of output	10

`--scale`	Scale to Int16, multiplying by 32,767	`False`

Any annual collections can be used as input for further export — for example, ESA WorldCover 10m v100 or ESA WorldCereal Active Cropland 10 m v100. Datasets with finer temporal granularity than annual will not be correctly aggregated by this tool.

Appendix B: Detailed Runtime Benchmarking Tables

Full GEE pipeline runtime and EECU time by year:

Year	Pipeline runtime (s)	Pipeline runtime (h)	EECU time (s)	EECU time (h)	Total size (GB)
2024	27,420.6247	7.6168	278,886.6827	77.4685	258.24
2023	30,263.8161	8.4066	264,226.6384	73.3962	258.64
2022	25,515.9630	7.0877	257,439.4404	71.5109	258.64
2021	27,541.3252	7.6504	262,592.8037	72.9424	258.48
2020	24,354.6962	6.7652	275,837.1997	76.6214	258.76
2019	41,363.0000	11.4892	302,880.4143	84.1334	258.11
2018	32,771.5112	9.1032	262,321.2517	72.8670	258.44
2017	42,716.2087	11.8656	287,451.6494	79.8477	260.33
AVERAGE	31,493.3931	8.7482	273,954.51	76.0985	258.705
TOTAL	251,947.145	**69.9853**	2,191,636.08	**608.7877**	**2.02 TiB**

Notes:

- Pipeline runtime is client wall-clock time, while EECU time reflects billable compute.
- EECU variability factor is up to 2.8× for identical requests.
- Total runtime variability factor is up to 8.8×.
- LSOA aggregation runtime is ~1.5 hours per year (8 CPUs, 32 GB RAM, GCS-mounted storage).

Appendix C: Full Validation Results and Supplementary Statistics

Range Validation Summary

All 24 million values (46,844 LSOAs × 64 dimensions × 8 years) passed the [−1, 1] range check with zero out-of-range values.

Random Sampling Validation — Full Results

Metric	Value
Total comparisons completed	100
Unique LSOAs sampled	42
Years covered	2017–2024
Mean difference (Pipeline – EE)	−0.000049
Std deviation of differences	0.000698
Mean correlation	0.9973
Min correlation	0.9859
Max correlation	1.0000
t-statistic	−0.7085
p-value	0.4803
Systematic bias detected	No

Yearly Breakdown

Year	Count	Mean Difference	Std Deviation
2017	11	−0.000027	0.000713
2018	13	−0.000176	0.000716
2019	13	0.000040	0.000887
2020	15	0.000082	0.000602
2021	10	0.000120	0.000648
2022	10	−0.000426	0.000538
2023	15	−0.000018	0.000821
2024	13	−0.000062	0.000605

Appendix D: Complete Output Schema and Field Descriptions

The output GeoPackage contains the following fields per row (one row = one LSOA × one year):

Field	Type	Description
data_zone_code	String	ONS LSOA (Lower layer Super Output Area) code for England and Wales, or equivalent data zone code for Scotland and Northern Ireland, representing the spatial area at which cloud probability is aggregated.
Geometry	MultiPolygon	MultiPolygon geometry representing the LSOA / data zone boundaries, stored in WKBGeometry format.
`band1_mean` … `band64_mean`	Float64	Mean embedding value per dimension across all valid pixels in the LSOA

Additional fields (median, min, max, std, sum per dimension) can be produced by the pipeline on request. Embedding values are in the [–1, 1] range (after descaling if the Int16 intermediate format is used).

> Although 2017 featured data quality issues (see here), they have been addressed in the latest version of the original Embedding data collection available on Google Earth Engine. The issue was related to dropout of some Sentinel-1 images, but the overall impact on dataset accuracy was quite low. As a rule of thumb, Google Earth Engine data is a source of truth and always contains the latest and updated version.

Appendix E: Scaling Precision Analysis (Including Error Statistics)

The Int16 scaling operation maps float64 values in [–1, +1] to integers in [–32767, +32767] via:

$$x_{\text{int16}} = \text{round}(x_{\text{float64}} \times 32767)$$

To recover original values:

$$x_{\text{original}} = \frac{x_{\text{int16}}}{32767}$$

Imago scaling:

Tile	Band	Max_err	Mean_err	RMSE
SJ26-2024	1	0.000015	0.000008	0.000009

SJ26-2024	2	0.000015	0.000007	0.000009
SJ26-2024	3	0.000015	0.000008	0.000009
SJ26-2024	4	0.000015	0.000007	0.000008
SJ26-2024	5	0.000015	0.000007	0.000008

Source Cooperative scaling:

Tile	Band	Max_err	Mean_err	RMSE
S084-2024	1	0.090058	0.000044	0.000898
S084-2024	2	0.104760	0.000053	0.001083
S084-2024	3	0.121553	0.000062	0.001264
S084-2024	4	0.109496	0.000054	0.001218
S084-2024	5	0.077017	0.000035	0.000713

Comparison

Statistic	IMAGO (Int16)	Source Cooperative (Int8)
Mean error	~10^{-6} (6th decimal place)	Higher
RMSE	~10^{-6} (6th decimal place)	Higher
Maximum absolute error	~10^{-5} (5th decimal place)	Higher

The IMAGO Int16 scaling preserves float64 values with a maximum error of sub-5th decimal place, which is negligible for all intended analytical applications. The Source Cooperative Int8 distribution, while more compact, has lower precision, particularly regarding maximum absolute error.

> Testing with GEE export in verbose mode revealed that the original Embedding tiling differs from the published Source Cooperative tile boundaries. For example, an area of interest intersecting four Source Cooperative tiles may overlay only two original Google Satellite Embedding images, while the output remains consistent and covers the entire area of interest.

See Figure 3 in the main text for visual comparison of scaling error distributions